Adaptation-Aware Model Selection for Few-Shot Digital Twin Instantiation under Deployment Limits

This supplementary material provides additional implementation, analytical, reproducibility, and diagnostic details for the main manuscript, including counting conventions, candidate-bank and calibration records, analytical derivations, baseline configurations, hardware profiling, external-transfer results, and synthetic-benchmark generation settings.

The source, architecture-screening, margin-calibration, diagnostic, and held-out target pools use disjoint center identities.

S1 Detailed Operation-Count and Parameter-Count Rules

The estimated operation count is computed for one forward pass with batch size one. Additions and multiplications in the main linear, convolutional, recurrent-matrix, residual-projection, and output-layer operations are counted. Bias additions, nonlinear activations, dropout operations, and GRU elementwise operations are excluded. The same deterministic rule is applied to every candidate.

For an instantiated architecture a_q, the parameter count is the exact number of elements in all trainable parameter tensors after the target input dimension and prediction horizon are fixed. Both counts depend on the target case only through its instantiated input and output dimensions. Because parameter adaptation does not change the architecture, both quantities remain unchanged.

The two indicators are constrained by hard deployment limits. They are not added to the prediction loss and are not changed after target adaptation. They are deterministic model-level indicators rather than direct guarantees of hardware-specific latency or memory use. Their relation to measured inference latency and serialized model size is evaluated in Supplementary Section S5.

S2 Candidate-Bank Records and Relative-Margin Calibration

S2.1 Candidate-bank construction and frozen records

The initial architecture space contains 66 indexed configurations: 18 MLPs, 36 TCNs, and 12 GRUs. Configuration A57, a one-layer GRU with 32 hidden units, is fixed as the reference architecture before screening.

An alternative configuration is retained when it records at least two check-oracle wins over A57, or at least two validation-selected positive check wins over A57. A check-oracle win is assigned to the feasible configuration with the lowest check loss when it improves on A57; a validation-selected positive check win is assigned when validation selects an alternative configuration whose check loss improves on A57. These cases belong to the architecture-screening pool and are disjoint from the margin-calibration and held-out target pools. The retained list contains three MLP and three GRU configurations. No TCN configuration satisfies this rule.

Two retained one-layer GRU-32 configurations have different indexed settings, but recurrent dropout is inactive for a one-layer GRU. They therefore represent the same executable architecture. The six retained configurations correspond to five executable architectures.

The final bank contains the frozen pooled-source PT+FT initialization for the reference, one source-trained initialization for each of the five other retained indexed configurations, and a second A57 initialization used as a paired-initialization control. The six retained configurations therefore produce seven candidates. The bank and reference candidate are frozen before relative-margin calibration and held-out evaluation.

S2.2 Relative-margin calibration records

Section 4.5 of the main manuscript defines the calibration rule. This section records its complete experimental instantiation and reproducibility details.

Let $\mathcal{T} = \{0.05, 0.075, 0.10, 0.125, 0.15, 0.20\}$ denote the preset margin grid. For a relative margin τ, let $\widehat{R}_{\text{harm}}(\tau)$ denote the observed harmful-selection rate. Let $\widehat{G}_{\text{ref}}(\tau)$ and $\widehat{G}_{\text{pair}}(\tau)$ denote the mean check-loss gains over the adapted reference and the paired-initialization control. Their lower 95% confidence bounds are $\underline{G}_{\text{ref}}(\tau)$ and $\underline{G}_{\text{pair}}(\tau)$.

The eligible margins form

$$\begin{aligned}\mathcal{T}_{\text{elig}} = \big\{\tau \in \mathcal{T} \;\big|\; & \widehat{R}_{\text{harm}}(\tau) \le \rho, \\ & \widehat{G}_{\text{ref}}(\tau) \ge \gamma_{\text{ref}}, \quad \underline{G}_{\text{ref}}(\tau) > 0, \\ & \widehat{G}_{\text{pair}}(\tau) \ge \gamma_{\text{pair}}, \quad \underline{G}_{\text{pair}}(\tau) > 0\big\}.\end{aligned} \tag{S1}$$

The calibration uses $\rho = 0.05$ and $\gamma_{\text{ref}} = \gamma_{\text{pair}} = 0.03$. When $\mathcal{T}_{\text{elig}}$ is nonempty, the calibrated margin is

$$\tau^* = \min \mathcal{T}_{\text{elig}}. \tag{S2}$$

The smallest eligible value in the main margin-calibration setting is $\tau^* = 0.10$.

Architecture screening and candidate-bank construction are completed before margin calibration. The candidate bank and reference candidate are fixed when the margin grid is evaluated.

If no margin satisfies the calibration criteria in a new domain, the adapted reference is retained. A margin fixed in another domain may still be evaluated without recalibration to assess transfer. The released calibration CSV contains the complete six-margin grid, harmful-selection rates, mean gains, and confidence bounds.

S3 Additional Analytical Details

S3.1 Ideal Candidate Decision within the Adapted Feasible Set

Let the expected future loss of adapted feasible candidate q on target center c be

$$R_c^{\text{cand}}(q) = R_c\left(a_q, \theta_{c,q}^{(T)}\right). \tag{S3}$$

If this quantity were available, the ideal candidate-level decision would satisfy

$$q_c^{\dagger} \in \operatorname*{arg\,min}_{q \in \mathcal{Q}_c^{\text{fea}}} R_c^{\text{cand}}(q). \tag{S4}$$

This candidate-level quantity is unavailable during instantiation. MSA-DTI therefore uses post-adaptation validation loss $J_c(q)$ together with the relative-margin rule in the main manuscript.

S3.2 Conditional expected-loss result

Assume that validation loss differs from expected future loss by at most $\epsilon_c \ge 0$ for every feasible candidate:

$$\left| R_c^{\text{cand}}(q) - J_c(q) \right| \le \epsilon_c, \qquad \forall q \in \mathcal{Q}_c^{\text{fea}}. \tag{S5}$$

When MSA-DTI selects an alternative, the relative-margin rule gives

$$R_c^{\text{cand}}(q_c^*) - R_c^{\text{cand}}(q^{\text{ref}}) \le -\tau J_c(q^{\text{ref}}) + 2\epsilon_c. \tag{S6}$$

Hence, the selected alternative has lower expected future loss than the reference when

$$\tau J_c(q^{\text{ref}}) > 2\epsilon_c. \tag{S7}$$

Equation (S7) is a sufficient condition under assumption (S5). The present experiments do not estimate an empirical bound on ϵ_c.

S4 Additional Experimental and Diagnostic Results

S4.1 Target-side baseline settings

Table S1 reports the target-side settings used in the main comparison.

Table S1: Target-side settings of MSA-DTI and the baselines in the main comparison.

Method	Structure	Source initialization	Target update	Adapted models	Deployment-limit check
PT+FT	Fixed reference	Pooled-source model	SGD/MSE–50	1	Before output
MeDeT-based adaptation	Fixed reference	Meta-learned initialization	SGD/MSE–50	1	Before output
Random initialization	Fixed reference	Random	SGD/MSE–50	1	Before output
Few-shot NAS	Top-12 shortlist	Method-specific models	Adam/Huber–50	≤ 12	Before evaluation
Zero-shot NAS	Top-ranked model	Method-specific model	None	0	Before evaluation
Zero-shot NAS + fine-tuning	Top-12 shortlist	Method-specific models	Adam/Huber–50	≤ 12	Before evaluation
H-Meta-NAS-based adaptation	Target architecture search	Architecture-indexed meta-initializations	SGD/MSE–50	Multiple candidates	Before search evaluation
MSA-DTI	Candidate bank	Frozen source initializations	SGD/MSE–50	4–7	Before adaptation and output

PT+FT and MSA-DTI use the same source-trained reference and target adaptation procedure. H-Meta-NAS-based adaptation uses the 50-update budget-matched condition in the main comparison. It uses the same 66 architecture definitions, filters infeasible architectures before search, uses source-only first-order meta-initializations, and selects the final architecture using target validation data. Its settings are fixed without held-out test data. All methods use the same target cases, deployment limits, and architecture definitions.

S4.2 Paired held-out baseline comparisons

Table S2 summarizes the paired MSE comparisons between MSA-DTI and the main baselines on the 80 held-out cases.

Table S2: Paired MSE comparisons between MSA-DTI and the main baselines on the held-out cases.

Baseline	MSE win rate (%)	Mean paired MSE reduction (%)	95% CI (%)
PT+FT	71.25	14.60	$[9.41, 20.14]$
MeDeT-based adaptation	87.50	35.37	$[25.39, 44.26]$
Random initialization	100.00	80.71	$[78.07, 83.22]$
Few-shot NAS	95.00	40.40	$[32.65, 47.58]$
Zero-shot NAS	100.00	76.29	$[68.95, 82.65]$
Zero-shot NAS + fine-tuning	98.75	45.03	$[36.53, 52.55]$
H-Meta-NAS-based adaptation	100.00	68.94	$[64.12, 73.43]$

MSE win rate is the percentage of paired cases in which MSA-DTI has lower test MSE. Positive reductions favor MSA-DTI. The 95% intervals are the 2.5th and 97.5th percentiles of 4000 center-cluster bootstrap replicates, with the four cases of each sampled target center kept together. The released evaluator derives the bootstrap seed deterministically from the frozen training seed, method name, and metric.

S4.3 Optimizer-matched diagnostic comparison

This comparison examines whether the performance ordering changes after the target optimizer, task loss, and update budget are matched. All methods use the same independent diagnostic target cases and the same deployment limits. The two common-shortlist controls use a shared 12-candidate set containing A57 and the 11 highest-ranked feasible alternatives under a fixed fused ranking (0.60 source-prior rank + 0.40 proxy rank). They differ only in the final selection rule: lowest validation loss versus the frozen 10% reference-based margin. All selections are fixed before test evaluation.

Table S3: Full optimizer-matched comparison on the diagnostic target pool.

Method	MSE↓	Worst-10%↓	CVaR90↓	Adapted models
MSA-DTI	**0.003809**	**0.015094**	**0.008696**	6.10
PT+FT	0.004497	0.017144	0.011878	1.00
Few-shot NAS	0.006715	0.025246	0.019653	12.00
Zero-shot NAS + fine-tuning	0.007402	0.028099	0.020402	12.00
Common shortlist, lowest validation loss	0.006707	0.025240	0.019653	12.00
Common shortlist, reference-based selection	0.006819	0.025562	0.020547	12.00

MSA-DTI gives the lowest MSE, Worst-10% error, and CVaR90 in this comparison. The diagnostic matches the target optimizer, task loss, and update budget while retaining each method's source initialization and selection rule.

S4.4 Selection-outcome details

On the 80 independent diagnostic cases, the relative-margin rule selects alternatives in 61 cases, including 49 beneficial and 12 harmful selections. Direct lowest-validation-loss selection chooses alternatives in 69 cases, including 55 beneficial and 14 harmful selections. The corresponding all-case harmful-selection rates are 15.00% and 17.50%, respectively.

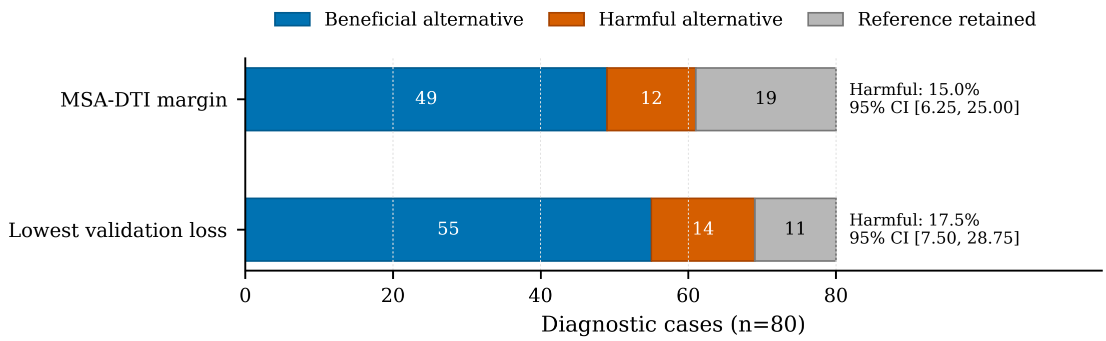

Figure S1: Selection outcomes on the 80-case diagnostic pool.

The relative-margin rule retains the reference more often. The confidence intervals of the two harmful-selection rates overlap, so the observed difference is not statistically confirmed.

S5 Timing, Hardware Profiling, and Deployment Analysis

S5.1 Target-side timing protocol

All methods are implemented in Python and PyTorch and run on an NVIDIA RTX 3060 Laptop GPU. Target-side time includes model creation, initialization loading, target-side search when applicable, target adaptation, validation evaluation, and final selection. It excludes offline preparation and test evaluation. Each reported time is the mean over five runs, with the corresponding standard deviation reported in the main manuscript.

S5.2 Architecture-profiling protocol

The five executable architectures are profiled with batch size one on an Intel Core i7-12700H CPU and an NVIDIA RTX 3060 Laptop GPU for $H \in \{1, 4\}$. Each setting uses 100 warm-up inferences followed by five repetitions of 1000 timed inferences. CUDA execution is synchronized before and after each timed region. Median latency is the primary latency measure, and the 95th percentile describes measurement variability. Serialized parameter-file size is used as the measured model size. Peak allocated GPU memory is also recorded for each architecture–horizon pair.

Table S4: Per-architecture batch-one hardware profile.

Configuration	H	Operations	Parameters	CPU (ms)	GPU (ms)	State (KiB)	GPU peak (KiB)
A1	1	155,712	77,921	0.097	0.148	306.5	1024
A1	4	155,904	78,020	0.094	0.133	306.9	1024
A6	1	157,760	78,977	0.093	0.168	311.0	1024
A6	4	157,952	79,076	0.099	0.151	311.4	1024
A13	1	159,808	80,033	0.128	0.237	315.5	1024
A13	4	160,000	80,132	0.128	0.218	315.9	1024
A55	1	377,888	2,081	1.793	3.999	10.3	54
A55	4	377,984	2,132	1.793	4.004	10.4	1030
A56/A57	1	1,050,688	5,697	1.980	4.019	24.4	60
A56/A57	4	1,050,880	5,796	1.975	4.170	24.8	1037

CPU and GPU values are median inference latency. A56 and A57 are the same executable one-layer GRU-32 architecture. GPU peak is the maximum allocated device memory recorded by the profiling protocol and is descriptive rather than a configured memory limit.

S5.3 Profiling results

Duplicate one-layer GRU-32 records are merged at the executable-architecture level. Across the CPU/GPU measurements and $H \in \{1, 4\}$, the Spearman correlation between estimated operation count and median inference latency ranges from 0.889 to 0.964. The correlation between parameter count and serialized parameter-file size is 1.000 in every measured setting, supporting parameter count as a model-size indicator. Peak GPU allocation is not monotonic in parameter count across the two horizons, so parameter count is not a direct proxy for runtime GPU memory. The MLP architectures use fewer estimated operations but more parameters than the GRU architectures. The two deployment indicators therefore capture different model properties, while their activity depends on the candidate set and limit tier. The profiling results characterize only the measured CPU/GPU host.

S5.4 Single-limit feasible-space analysis

To identify the active constraint in the frozen candidate bank, we apply the operation-count and parameter-count limits separately to the same seven candidates using their fixed complexity profiles.

For both $H = 1$ and $H = 4$, all seven candidates satisfy the operation-count limit under the tight, medium, and loose settings. Under the tight parameter-count limit, the three MLP candidates are excluded and the feasible bank is reduced from seven candidates to four, whereas the medium and loose parameter-count limits retain all seven candidates. Consequently, the joint feasible set contains four candidates under the tight setting and seven under the medium and loose settings. The fixed reference candidate satisfies both limits in all six horizon–tier combinations.

S5.5 Expanded-bank dual-limit stress analysis

The preceding analysis shows that the operation-count limit is inactive for the frozen seven-candidate main bank under the main limit tiers. To examine whether operation count and parameter count can independently restrict candidate feasibility, we additionally construct an eight-candidate stress bank by adding configuration 24 from the predefined architecture space.

The stress thresholds are derived solely from deterministic model-complexity profiles. Across $H \in \{1, 4\}$, the largest operation count in the original seven-candidate bank is 1,050,880, whereas the smallest operation count of configuration 24 is 1,198,112. Their midpoint gives the stress operation-count limit $B^F = 1{,}124{,}496$. Similarly, the largest parameter count among the stress-bank candidates other than A13 is 79,076, whereas the smallest parameter count of A13 is 80,033. Their midpoint gives the stress parameter-count limit $B^P = 79{,}554.5$.

For both $H = 1$ and $H = 4$, configuration 24 violates only the operation-count limit, A13 violates only the parameter-count limit, and the fixed reference A57 satisfies both limits. Each single-limit feasible set therefore retains seven of the eight candidates, whereas their joint feasible set retains six. Thus, the joint feasible set is a strict subset of each single-limit feasible set.

The target-side stress evaluation uses the same 80 held-out cases, and all models selected under the joint limits satisfy both limits. The expanded bank is used only for this limit-complementarity diagnostic, whereas the main evaluation uses the original seven-candidate bank.

S5.6 Selected-architecture analysis

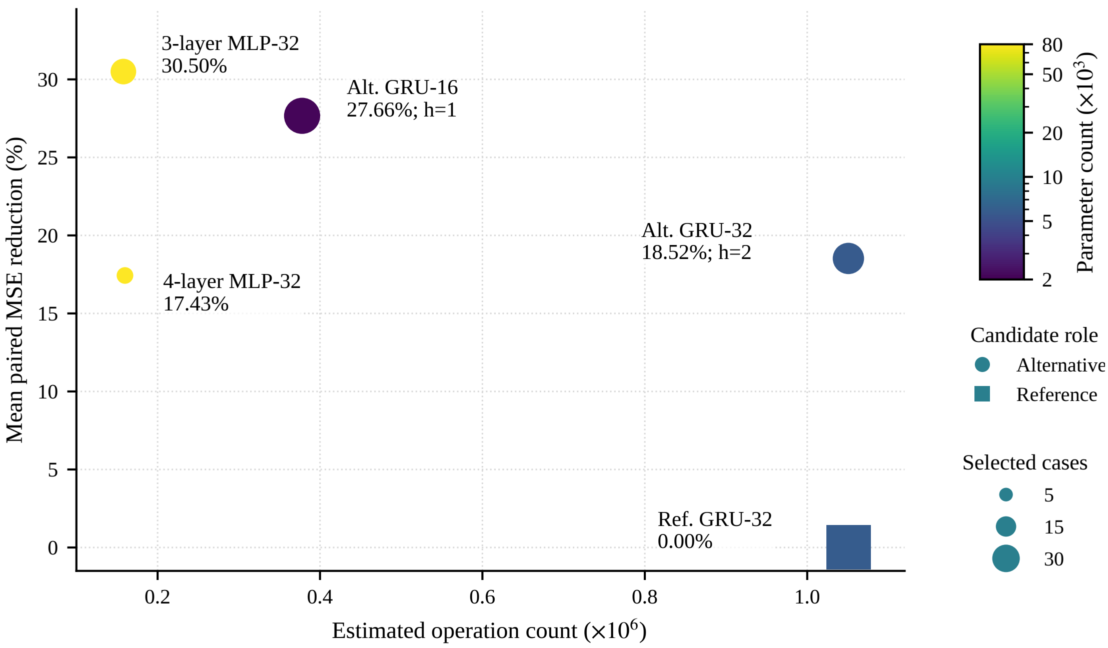

Figure S2: Selected-configuration complexity and performance. Marker area shows the number of selected cases, and h denotes harmful selections.

The reference configuration is retained in 33 cases. Four alternative configurations cover the remaining 47 cases, and their mean paired MSE reductions range from 17.43% to 30.50%. All four alternatives have positive mean reductions. The three harmful selections occur on two alternative GRU configurations.

S6 Configuration Analysis

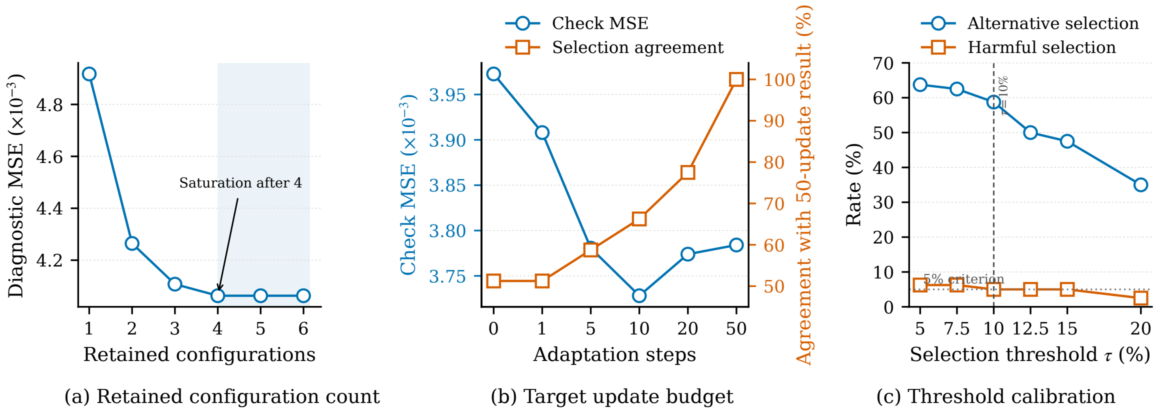

Figure S3: Configuration analysis of (a) the retained configuration count, (b) the target update budget, and (c) the relative margin.

Diagnostic MSE decreases from 0.004917 with one retained configuration to 0.004063 with four retained configurations and then remains unchanged. The main evaluation uses the six-configuration list fixed before evaluation. The observed saturation after four configurations supports a compact bank, while this diagnostic does not optimize the bank size itself.

Most of the observed reduction in diagnostic check MSE occurs within the first ten target updates. The main experiment nevertheless uses 50 updates for every feasible candidate to preserve a common fixed target budget.

Using the same frozen candidate bank, source initializations, feasibility rule, and reference-based selection rule at both endpoints, selected check MSE is 0.003973 with zero target updates and 0.003784 after 50 updates. The mean paired reduction is 4.75%, with a 95% confidence interval of $[-1.62\%, 10.23\%]$, and the same candidate is selected at both endpoints in 51.25% of cases. The interval spans zero, so the isolated performance difference remains uncertain in this diagnostic.

The relative-margin analysis shows that 10% is the smallest value satisfying all preset calibration criteria.

S7 Source Sensitivity and External Transfer

S7.1 Source-center scale and source-training seeds

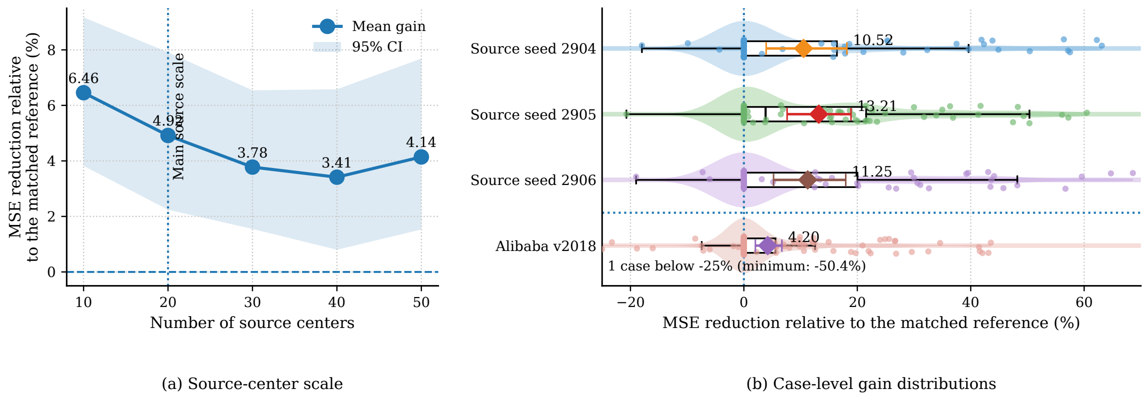

Figure S4: Sensitivity to source-center scale, source-training seeds, and Alibaba v2018 transfer.

One Alibaba case below -25% is clipped in Fig. S4 only for visualization; all original values are retained in the analysis.

The mean paired MSE reductions relative to the matched reference at source-center scales 10, 20, 30, 40, and 50 are 6.46%, 4.92%, 3.78%, 3.41%, and 4.14%, respectively. The mean gain remains positive at every tested source scale.

Across source-training seeds 2904, 2905, and 2906, the mean paired MSE reductions are 10.52%, 13.21%, and 11.25%. All three confidence intervals remain above zero, although some individual cases have zero or negative gains. The screened architecture list is held fixed in this seed study. The result therefore measures sensitivity to source-model training and initialization.

S7.2 Independent architecture-screening stability

We separately repeat the complete 66-configuration diagnostic screening on two new 20-center pools. Together with the original 20-center screening pool, this gives three independently sampled screening sets. The A57 reference, architecture space, initialization recipe, SGD/MSE–50 target recipe, and the screening and retention rules defined in Supplementary Section S2.1 are kept unchanged. None of these pools overlaps the 20-center robustness pool used below, and no test data are used during screening.

Table S5: Independent configuration-screening stability and evaluation on centers 1160–1179.

Screen	Centers	Retained configurations	Size	Jaccard	Gain (%)	95% CI (%)	Harmful (%)
Frozen S0	900–919	A1, A6, A13, A55, A56, A57	6	1.000	0.295	$[-0.646, 1.564]$	2.50
Independent S1	1120–1139	reference fallback only	0	0.000	0.000	$[0.000, 0.000]$	0.00
Independent S2	1140–1159	A55, A56, A57, A59	4	0.429	0.125	$[-0.894, 1.427]$	3.75

Jaccard similarity is relative to the frozen S0 list. Size counts retained configurations before adding the protected PT-A57 fallback. Gain is the mean paired check-MSE reduction relative to PT-A57 on 80 robustness cases; intervals use a 4000-repetition center-cluster bootstrap. The diagnostic evaluates screening stability without retraining or retuning the main evaluation protocol.

Across S0–S2, retained-list sizes are 6, 0, and 4 (median 4, range 0–6). A55, A56, and A57 are retained in two of three pools; A1, A6, A13, and A59 are each retained once. On the robustness pool, the S0 list selects PT-A57 in 72 cases, A55 in three, A56 in three, and the paired-control A57 initialization in two. S1 falls back to PT-A57 in all 80 cases. S2 selects PT-A57 in 71 cases, A55 in three, A56 in three, the paired-control A57 initialization in two, and A59 in one.

The two nonempty retained lists, S0 and S2, both have paired-gain intervals that include zero on the robustness pool, indicating substantial sample sensitivity. Accordingly, the main evaluation uses the frozen S0 bank, while the persistent A57 reference provides a fallback when a screening pool retains no alternative.

Reapplying the recorded screening counts with retention thresholds of one, two, and three wins retains 10, 6, and 2 configurations, respectively. The frozen threshold-two list is S0, whereas threshold three retains only A55 and A57. On the existing step-50 diagnostic trajectory, threshold three increases selected check MSE by 3.79% relative to threshold two, with a 95% confidence interval of $[1.58\%, 6.30\%]$. The threshold-one list additionally contains A4, A5, A12, and A59, for which source-trained initializations under the frozen bank-construction protocol are unavailable; downstream results are therefore reported only for thresholds two and three.

S7.3 Alibaba external-domain transfer

Alibaba Cluster Trace v2018 is split into 20 source machines, 20 calibration machines, and 40 held-out machines. The calibration and held-out pools contain 80 and 160 cases, respectively. Models are retrained on the Alibaba source machines using their own architectures and source-only normalization. The architecture definitions, deployment filtering rule, target protocol, and margin grid remain unchanged.

The workload observations come from the production trace, while the deployment-limit tiers reuse the controlled model-complexity settings from the main study. This setting isolates external workload-domain transfer under the frozen deployment protocol.

No margin in the fixed grid satisfies all Alibaba calibration criteria. Under the calibration rule, the adapted reference is therefore retained, while the fixed main-study margin $\tau = 0.10$ is reported separately to assess transfer without recalibration.

Under direct transfer of the fixed main-study margin $\tau = 0.10$, the adapted reference and the fixed-margin MSA-DTI diagnostic have mean MSE values of 0.001337 and 0.001279, respectively, corresponding to a 4.29% aggregate reduction. The mean paired MSE reduction is 4.20%, with a 95% confidence interval of $[2.03\%, 6.70\%]$. The fixed-margin diagnostic selects alternatives in 57 cases, including 47 beneficial and 10 harmful selections. The harmful-selection rates are 6.25% over all held-out cases and 17.54% conditional on alternative selection. Thus, direct transfer yields a positive average MSE signal, while the 5% calibration criterion is not met on Alibaba.

S8 Synthetic Benchmark Generation and Center Heterogeneity

The controlled benchmark represents heterogeneous computing centers in a common metric coordinate space. Each center may observe only a subset of the metric positions. The model input uses 12 value channels, 12 corresponding observation-mask channels, and one time-gap channel, giving a fixed input dimension of 25. Unavailable metric values are zero-filled and identified by the corresponding mask channels.

Center types A–C are evaluation strata rather than model classes. They differ in workload dynamics, sampling, and monitoring conditions. Deployment-limit tiers are separate platform-side settings and are assigned using the type-dependent probabilities reported in Table S6. The table also gives the complete type-specific parameter ranges. The default A:B:C ratio is 3:4:3, giving 6 Type-A, 8 Type-B, and 6 Type-C centers in each 20-center source or held-out target pool.

Table S6: Type-specific parameter ranges of the controlled synthetic benchmark.

Parameter	Type A	Type B	Type C
Sampling interval d_t	$\{1\}$	$\{1, 2\}$	$\{2, 3, 4\}$
Point-missing rate	$[0.02, 0.08]$	$[0.10, 0.25]$	$[0.25, 0.45]$
Missing-block length	$[5, 15]$	$[20, 60]$	$[60, 120]$
Observation-noise std.	$[0.01, 0.03]$	$[0.03, 0.06]$	$[0.06, 0.10]$
Permanently unavailable metrics	0–2	2–4	4–6
Base workload logit	$[-0.70, 0.10]$	$[-0.45, 0.35]$	$[-0.20, 0.55]$
Seasonal amplitude	$[0.20, 0.45]$	$[0.30, 0.60]$	$[0.40, 0.75]$
AR coefficient	$[0.55, 0.80]$	$[0.60, 0.88]$	$[0.65, 0.92]$
Process-noise std.	$[0.04, 0.08]$	$[0.06, 0.11]$	$[0.08, 0.14]$
Burst rate per 1000 base steps	$[1.0, 2.5]$	$[1.5, 3.5]$	$[2.0, 4.5]$
Burst amplitude	$[0.25, 0.55]$	$[0.35, 0.75]$	$[0.45, 0.95]$
Drift probability	0.20	0.40	0.60
Drift strength	$[0.10, 0.25]$	$[0.15, 0.35]$	$[0.20, 0.45]$
Limit-tier probabilities (tight / medium / loose)	0.25/0.45/0.30	0.35/0.45/0.20	0.45/0.40/0.15

The three types share the same missingness mechanisms. For every center, the generator may create 2–6 single-metric outage blocks and 1–3 grouped outage blocks containing 2–4 metrics. A short global monitoring outage occurs with probability 0.20. The shared periodic base scales are 24, 96, and 672 base steps.

Together, these settings vary five aspects of the benchmark: workload dynamics, sampling, monitored-variable availability, observation conditions, and deployment-limit settings.

For each center, the generator first creates a latent workload process. The workload contains periodic variation, autoregressive variation, and random workload bursts. When drift occurs, it is generated as either an abrupt change or a gradual transition.

The latent workload is then mapped to multivariate monitoring signals. Individual metrics may have different delays, gains, offsets, nonlinear terms, and observation noise. Center-specific sampling is applied before monitoring-schema and missingness effects are introduced.

The benchmark generator and all data-role assignments are fixed before architecture screening, margin calibration, diagnostics, and held-out evaluation.